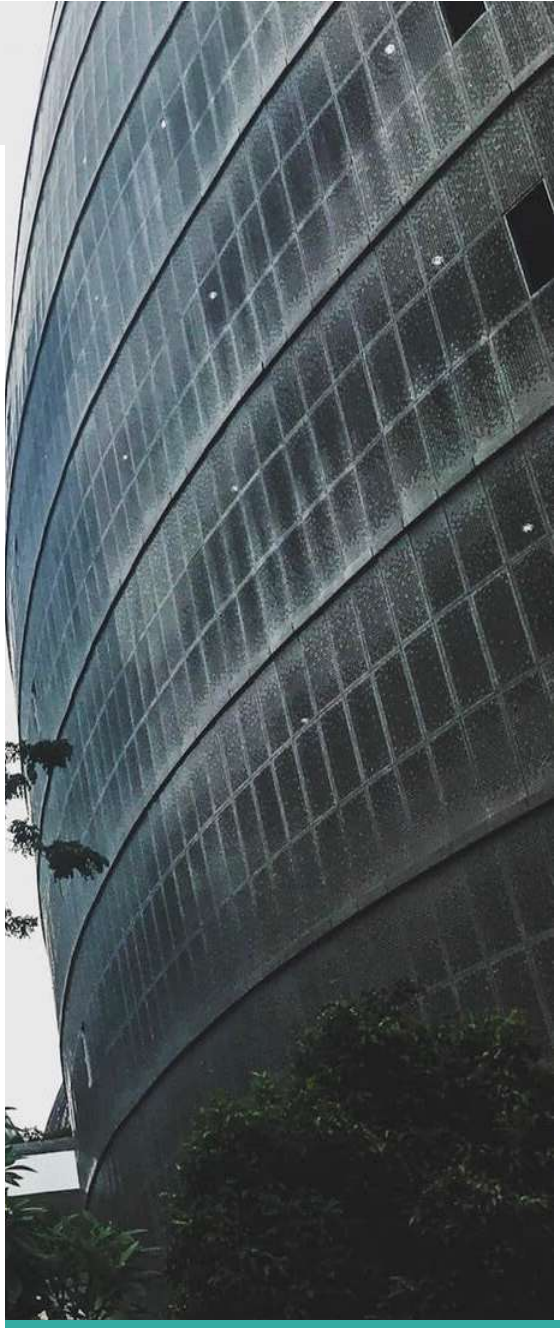
MODUL PEMBELAJARAN

IF670 – Mobile Cross Platform Programming
S1 INFORMATICS GRADUATE PROGRAM
FACULTY OF ENGINEERING AND INFORMATICS



PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
UNIVERSITAS MULTIMEDIA NUSANTARA

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MODUL 10
(Maps & Geolocation)

DESKRIPSI TEMA

Implementasi Geolocation dan Map View menggunakan OpenStreetMap di React Native Expo.

CAPAIAN PEMBELAJARAN MINGGUAN (SUB-CAPAIAN PEMBELAJARAN)

Mahasiswa mampu mengambil lokasi pengguna menggunakan GPS, menampilkan peta berbasis OpenStreetMap, dan menampilkan koordinat latitude dan longitude.

PENUNJANG PRAKTIKUM

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

LANGKAH-LANGKAH PRAKTIKUM

1. Install dependencies "npx expo install expo-location react-native-maps"
2. Tambahan Button "Get Geo Location" yang dimana Button tersebut akan meminta "Location Permission" untuk mengambil koordinat user
3. Menampilkan Map untuk mengambil Latitude dan Longitude

```
import * as Location from "expo-location";
import React, { useState } from "react";
import { Button, Dimensions, StyleSheet, Text, View } from "react-native";
import MapView, { Marker, Region, UrlTile } from "react-native-maps";

type Coordinates = {
  latitude: number;
  longitude: number;
};

// Get the device's screen height for responsive map sizing
const { height } = Dimensions.get("window");

export default function App() {
  const [location, setLocation] = useState<Coordinates | null>(null);

  //async function to get the user's current location
  const getLocation = async (): Promise<void> => {
    const { status } = await Location.requestForegroundPermissionsAsync();

    if (status !== "granted") {
      alert("Permission denied! Please allow location access.");
      return;
    }

    const loc = await Location.getCurrentPositionAsync({});
    setLocation({
      latitude: loc.coords.latitude,
      longitude: loc.coords.longitude,
    });
  };
```

```
33    // Define the region for the map based on the user's location
34    const region: Region | undefined = location
35      ? {
36          latitude: location.latitude,
37          longitude: location.longitude,
38          latitudeDelta: 0.01,
39          longitudeDelta: 0.01,
40        }
41      : undefined;
42
43    return (
44      <View style={styles.container}>
45        {!location ? (
46          <Button title="Get Geo Location" onPress={getLocation} />
47        ) : (
48          <>
49            <MapView style={styles.map} initialRegion={region}>
50              {/* OpenStreetMap Tile */}
51              <UrlTile urlTemplate="https://tile.openstreetmap.org/{z}/{x}/{y}.png" />
52
53              {/* Marker */}
54              <Marker coordinate={location} title="My Location" />
55            </MapView>
56
57            <View style={styles.info}>
58              <Text>Latitude: {location.latitude}</Text>
59              <Text>Longitude: {location.longitude}</Text>
60
61              <Button title="Refresh Location" onPress={getLocation} />
62            </View>
63          </>
64        )}
65      </View>
66    );
67  }
```

```
69  const styles = StyleSheet.create({
70    container: {
71      flex: 1,
72    },
73    center: {
74      flex: 1,
75      justifyContent: "center",
76      alignItems: "center",
77    },
78    map: {
79      height: height * 0.5,
80      width: "100%",
81    },
82    info: {
83      flex: 1,
84      padding: 16,
85      backgroundColor: "#fff",
86    },
87  });
```

TUGAS

Setelah kalian berhasil mendapatkan current location, kembangkan aplikasi dengan menambahkan fitur berikut:

- Buat marker pada peta dapat diubah posisinya
- Perubahan posisi marker dapat dilakukan dengan:
    - menekan (tap) pada peta, atau
    - menggeser (drag) marker

Setiap perubahan posisi marker harus langsung memperbarui nilai latitude dan longitude yang ditampilkan pada layar!

REFERENSI